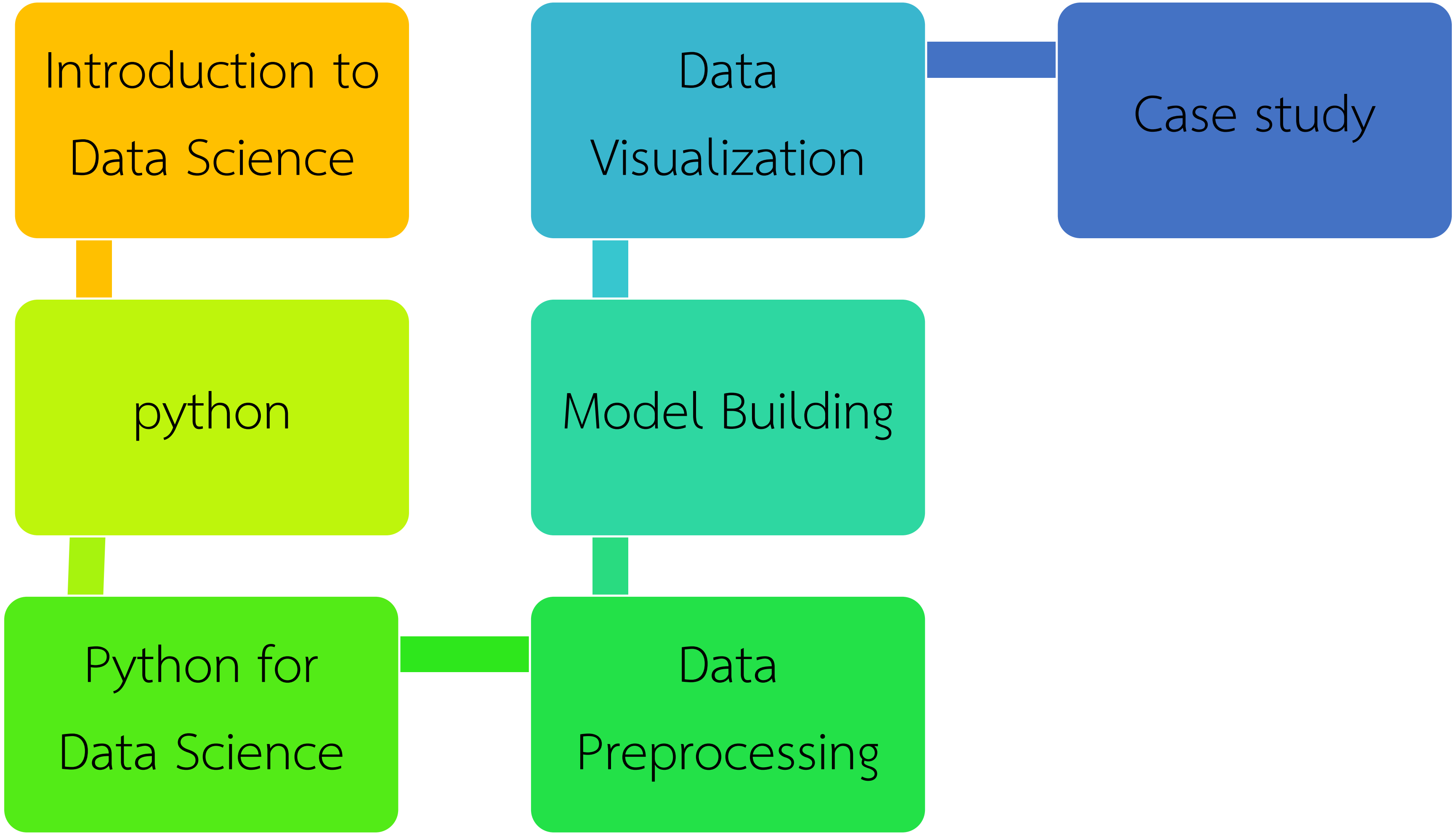
รหัสวิชา 2-233-225
ชื่อวิชา วิทยาศาสตร์ข้อมูลเบื้องต้น
(Introduction to Data Science)
สัปดาห์ที่ 11

CS

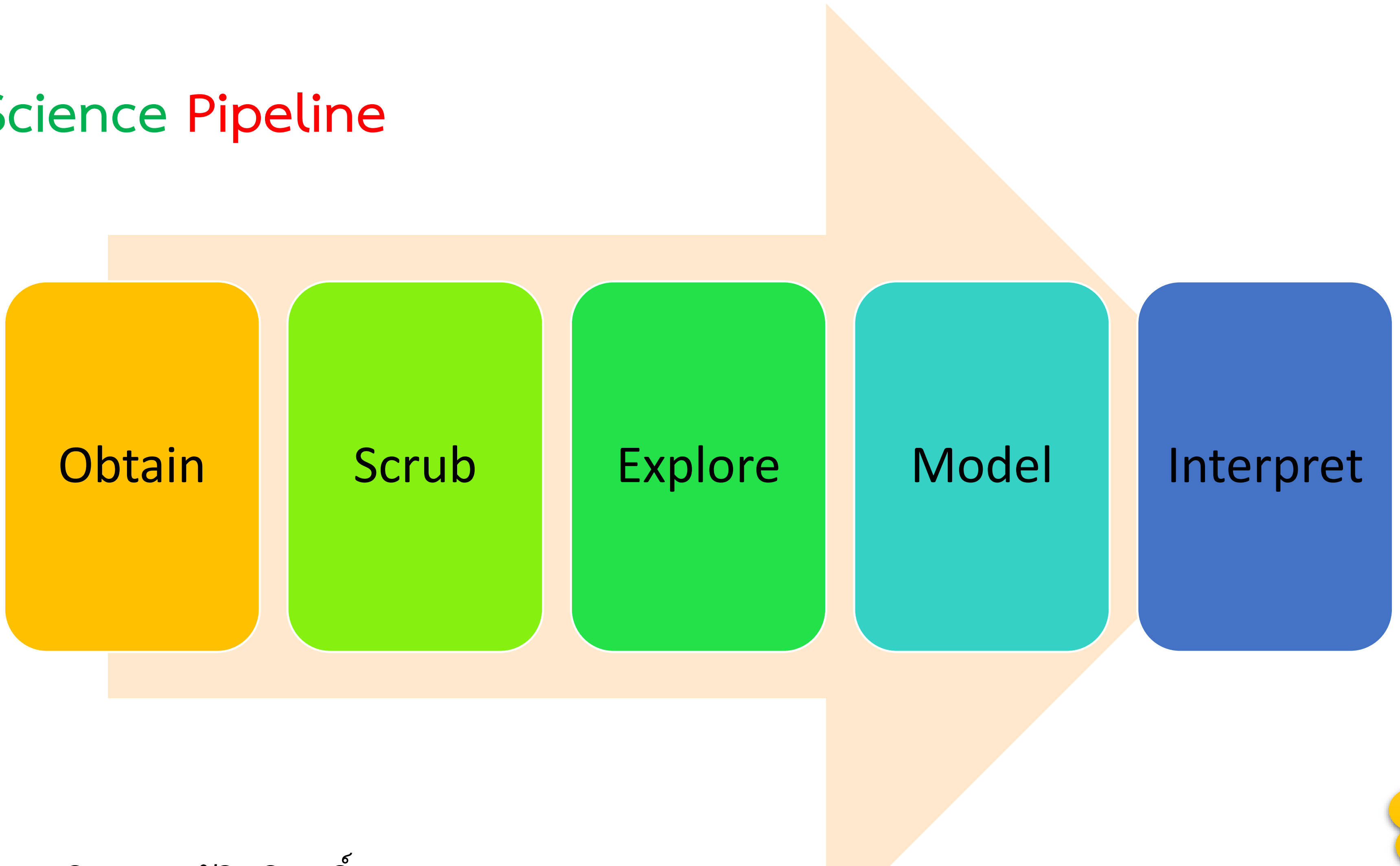
เนื้อหาของรายวิชา



CS

ขั้นตอนการทำ Data Science

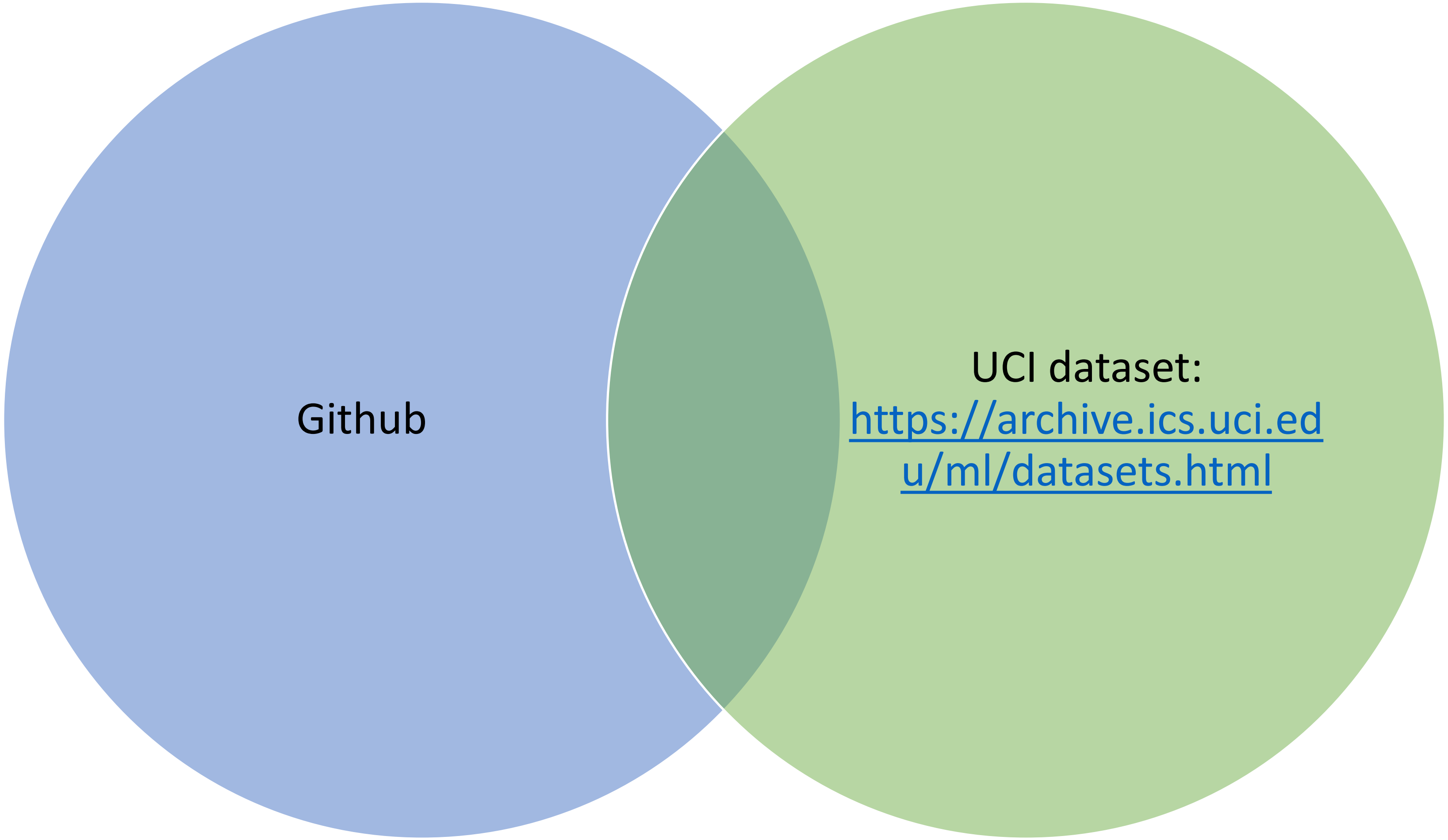
Data Science Pipeline



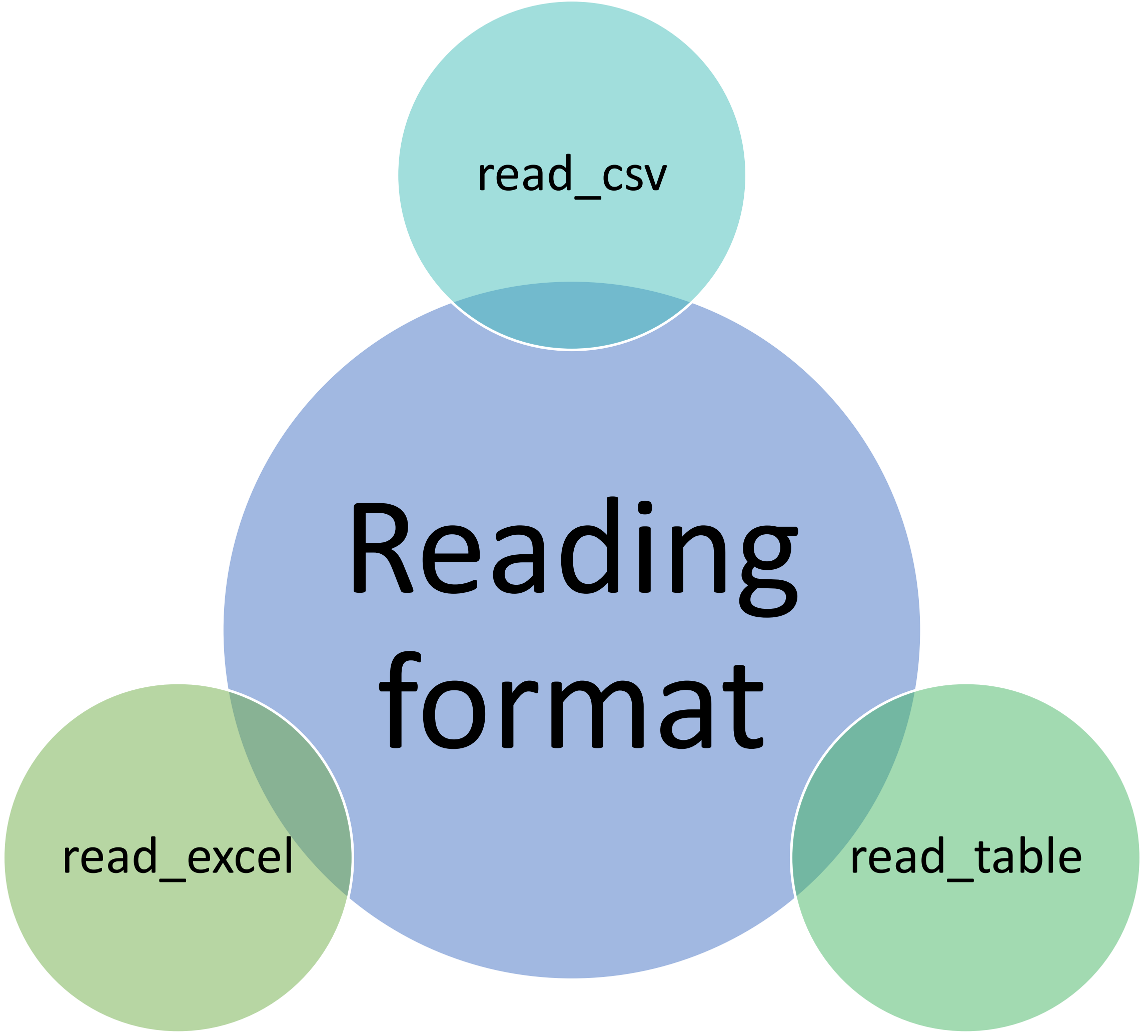
(อรพิน ประวัติบริสุทธิ์, 2564)

CS

Source: Dataset



https://github.com/awesomedata/awesome-public-datasets https://archive.ics.uci.edu/ml/index.php



[Jesús Rogel-Salazar, 2017]

Example: Read CSV file

```
In [3]: import pandas as pd
```

```
In [4]: url= "https://github.com/AddisonGauss/NbaData2015-2016/raw/master/nbasalaries.csv"
        df = pd.read_csv(url)
        df
```

Out[4]:

	Unnamed: 0	RK	Player	TEAM	SALARY
0	0	1	Kobe Bryant	Los Angeles Lakers	$25,000,000
1	1	2	LeBron James	Cleveland Cavaliers	$22,970,500
2	2	3	Carmelo Anthony	New York Knicks	$22,875,000
3	3	4	Dwight Howard	Houston Rockets	$22,359,364
4	4	5	Chris Bosh	Miami Heat	$22,192,730
5	5	6	Chris Paul	LA Clippers	$21,468,695
6	6	7	Kevin Durant	Oklahoma City Thunder	$20,158,622

https://github.com/AddisonGauss/NbaData2015-2016/blob/master/nbasalaries.csv

[Jesús Rogel-Salazar, 2017]

Example: Read TSV file

In [6]:
```
url='https://github.com/prasertcbs/test/raw/master/top_movies.tsv'
df =pd.read_table(url)
df
```

Out[6]:

	Rank	Title	Worldwide gross	Year
0	1	Avatar	$2,782,275,172	2009
1	2	Titanic film currently playing	$2,175,778,696	1997
2	3	Harry Potter and the Deathly Hallows – Part 2	$1,328,111,219	2011
3	4	Transformers: Dark of the Moon	$1,123,746,996	2011
4	5	The Lord of the Rings: The Return of the King	$1,119,929,521	2003
5	6	Pirates of the Caribbean: Dead Man's Chest	$1,066,179,725	2006
6	7	Toy Story 3	$1,063,171,911	2010
7	8	Pirates of the Caribbean: On Stranger Tides	$1,043,871,802	2011
8	9	Star Wars Episode I: The Phantom Menace film c...	$1,026,281,382	1999
9	10	Alice in Wonderland	$1,024,299,904	2010

https://github.com/prasertcbs/test/blob/master/top_movies.tsv

[Jesús Rogel-Salazar, 2017]

Example: Read Excel file

In [7]:
```
url='https://github.com/prasertcbs/test/raw/master/iris.xlsx'
df = pd.read_excel(url)
df
```

Out[7]:

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa
6	4.6	3.4	1.4	0.3	setosa
7	5.0	3.4	1.5	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa

https://github.com/prasertcbs/test/blob/master/iris.xlsx

[Jesús Rogel-Salazar, 2017]

การลบแถวและคอลัมน์

Drop row แบบชั่วคราว

- df.drop(X)

Drop row แบบถาวร

- df.drop(X, inplace=True, axis=0)

Drop column แบบชั่วคราว

- df.drop(['mpg'], axis=1)

Drop column แบบถาวร

- df.drop(['mpg'], axis=1, inplace=True)

การเปลี่ยนชื่อ column

- Rename(columns = {‘XXX’: ‘XX’}, inplace = True)

[Jesús Rogel-Salazar, 2017]

Drop row แบบชั่วคราว

```
In [1]: import pandas as pd
```

```
In [2]: url = 'https://github.com/prasertcbs/tutorial/raw/master/mtcars.csv'
        df = pd.read_csv(url)
        df.head()
```

Out[2]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
2	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2

```
In [3]: df.index

Out[3]: RangeIndex(start=0, stop=32, step=1)
```

```
In [4]: df.columns

Out[4]: Index(['model', 'mpg', 'cyl', 'disp', 'hp', 'drat', 'wt', 'qsec', 'vs', 'am',
               'gear', 'carb'],
              dtype='object')
```


[Jesús Rogel-Salazar, 2017]

https://github.com/prasertcbs/tutorial/raw/master/mtcars.csv

Drop row แบบชั่วคราว

In [5]:
```
df.drop(2)
```

Out[5]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2

In [6]:
```
df
```

Out[6]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
2	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2

[Jesús Rogel-Salazar, 2017]

Drop row แบบถาวร

```
In [7]: df.drop(2, inplace=True, axis=0)
```

```
In [8]: df
```

Out[8]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2

[Jesús Rogel-Salazar, 2017]

Drop column แบบชั่วคราว

In [9]:
```
df.drop(['mpg'], axis=1)
```

Out[9]:

	model	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	6	160.0	110	3.90	2.875	17.02	0	1	4	4
3	Hornet 4 Drive	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	Merc 240D	4	146.7	62	3.69	3.190	20.00	1	0	4	2

[Jesús Rogel-Salazar, 2017]

Drop column แบบถาวร

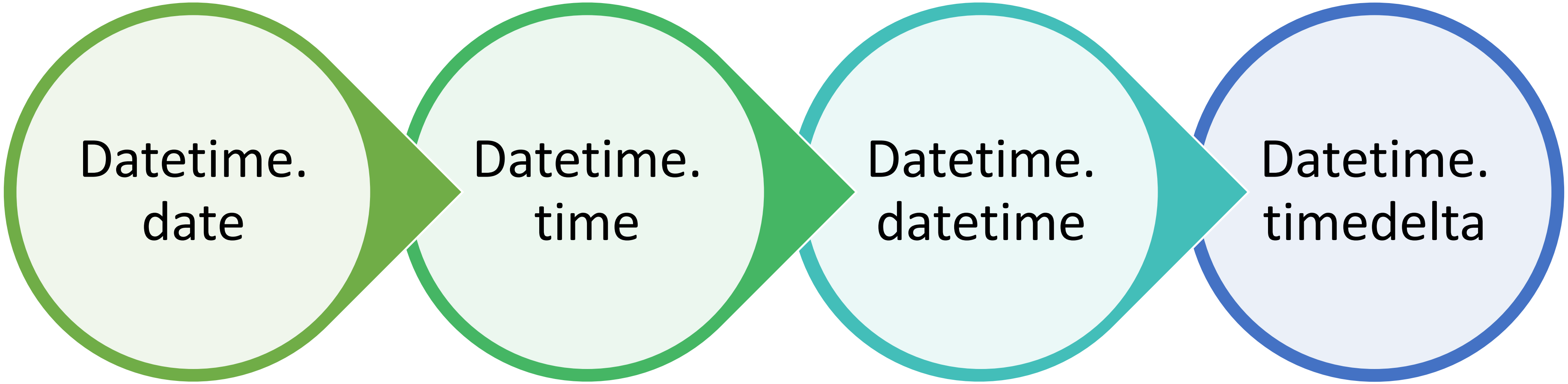
```
In [11]: df.drop(['mpg'], axis=1, inplace=True)
         df
```

Out[11]:

	model	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	6	160.0	110	3.90	2.875	17.02	0	1	4	4
3	Hornet 4 Drive	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	8	360.0	245	3.21	3.570	15.84	0	0	3	4

[Jesús Rogel-Salazar, 2017]

Dates and Times in Python



https://www.youtube.com/watch?v=ciXCePAEvxE

[Jesús Rogel-Salazar, 2017]

Dates and Times in Python

```
In [68]: from datetime import datetime
         datetime(year=2019, month = 9, day=2)

Out[68]: datetime.datetime(2019, 9, 2, 0, 0)
```

https://www.youtube.com/watch?v=ciXCePAEvxE

[Jesús Rogel-Salazar, 2017]

Dates and Times in Python

```
In [69]: from dateutil import parser
         date = parser.parse("2nd of September, 2019")
         date

Out[69]: datetime.datetime(2019, 9, 2, 0, 0)
```

https://www.youtube.com/watch?v=ciXCePAEvxE

[Jesús Rogel-Salazar, 2017]

Typed arrays of times: NumPy’s datetime64

In [10]:
```
import numpy as np
date = np.array('2019-09-02', dtype = np.datetime64)
date
```

Out[10]:
```
array('2019-09-02', dtype='datetime64[D]')
```

In [11]:
```
date+np.arange(12)
```

Out[11]:
```
array(['2019-09-02', '2019-09-03', '2019-09-04', '2019-09-05',
       '2019-09-06', '2019-09-07', '2019-09-08', '2019-09-09',
       '2019-09-10', '2019-09-11', '2019-09-12', '2019-09-13'],
      dtype='datetime64[D]')
```

https://www.youtube.com/watch?v=ciXCePAEvxE

[Jesús Rogel-Salazar, 2017]

การแสดงวันและเวลาในรูปแบบต่าง ๆ

```
In[6]: np.datetime64('2021-01-26')
Out[6]: numpy.datetime64(‘2021-01-26')
```

```
In[7]: np.datetime64('2021-01-26 12:00', 'm')
Out[7]: numpy.datetime64('2021-01-26T12:00')
```

```
In[8]: np.datetime64('2021-01-26 12:59:59.50', 'ns')
Out[8]: numpy.datetime64(' 2021-01-26 -04T12:59:59.500000000')
```

https://www.youtube.com/watch?v=ciXCePAEvxE

[Jesús Rogel-Salazar, 2017]

Converting Between String and Datetime

In [9]:
```
from datetime import datetime
stamp = datetime(2019,9,9)
```

In [10]:
```
str(stamp)
```

Out[10]:
```
'2019-09-09 00:00:00'
```

In [11]:
```
stamp.strftime('%Y-%m-%d')
```

Out[11]:
```
'2019-09-09'
```


[Jesús Rogel-Salazar, 2017]

Datetime format specification

%Y	Four-digit year
%y	Two-digit year
%m	Two-digit month [01, 12]
%d	Two-digit day [01, 31]
%H	Hour (24-hour clock) [00, 23]
%I	Hour (12-hour clock) [01, 12]
%M	Two-digit minute [00, 59]
%S	Second [00, 61] (seconds 60, 61 account for leap seconds)
%w	Weekday as integer [0 (Sunday), 6]

[Jesús Rogel-Salazar, 2017]

Datetime format specification

%U	Week number of the year [00, 53]; Sunday is considered the first day of the week, and days before the first Sunday of the year are "week 0"
%W	Week number of the year [00, 53]; Monday is considered the first day of the week, and days before the first Monday of the year are "week 0"
%z	UTC time zone offset as +HHMM or -HHMM; empty if time zone naive
%F	Shortcut for %Y-%m-%d (e.g., 2012-4-18)
%D	Shortcut for %m/%d/%y (e.g., 04/18/12)

[Jesús Rogel-Salazar, 2017]

การใช้ cut เพื่อแบ่งข้อมูลออกเป็น category (binning)

- สร้าง list มาเก็บ ช่วง bin ที่ต้องการ
- map ค่า list ที่สร้างไว้ไปใส่ใน Label และเรียงลำดับตาม list
- เรียกดูค่า attribute ที่เรียงลำดับแล้ว

[Jesús Rogel-Salazar, 2017]

binning

In [1]:
```
import pandas as pd
import numpy as np
%matplotlib inline
```

In [3]:
```
df=pd.read_csv('https://github.com/prasertcbs/tutorial/raw/master/staff.csv',
               index_col='empID',
               thousands=',', parse_dates=['dob', 'join_date'])
df
```

Out[3]:

empID	fname	lname	sex	dob	position	department	salary	join_date
604	กันตภณ	ซ่อนกลิ่น	M	1994-03-05	ผู้จัดการ	HR	13000	2016-02-28
607	เกศินี	สายหยุด	F	1968-12-06	ผู้ช่วยผู้จัดการ	FIN	40000	1987-11-30

https://github.com/prasertcbs/tutorial/raw/master/staff.csv

[Jesús Rogel-Salazar, 2017]

binning

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 12 entries, 604 to 724
Data columns (total 8 columns):
fname          12 non-null object
lname          12 non-null object
sex            12 non-null object
dob            12 non-null datetime64[ns]
position       12 non-null object
department     12 non-null object
salary         12 non-null int64
join_date      12 non-null datetime64[ns]
dtypes: datetime64[ns](2), int64(1), object(5)
memory usage: 864.0+ bytes
```

```
In [3]: min_value = df['salary'].min()
        max_value = df['salary'].max()
        print(min_value)
        print(max_value)
```

```
13000
101500
```


[Jesús Rogel-Salazar, 2017]

binning

In [4]:
```
import numpy as np
bins = [0, 15000, 30000, 50000, 80000,np.inf]
bins
```

Out[4]:
```
[0, 15000, 30000, 50000, 80000, inf]
```

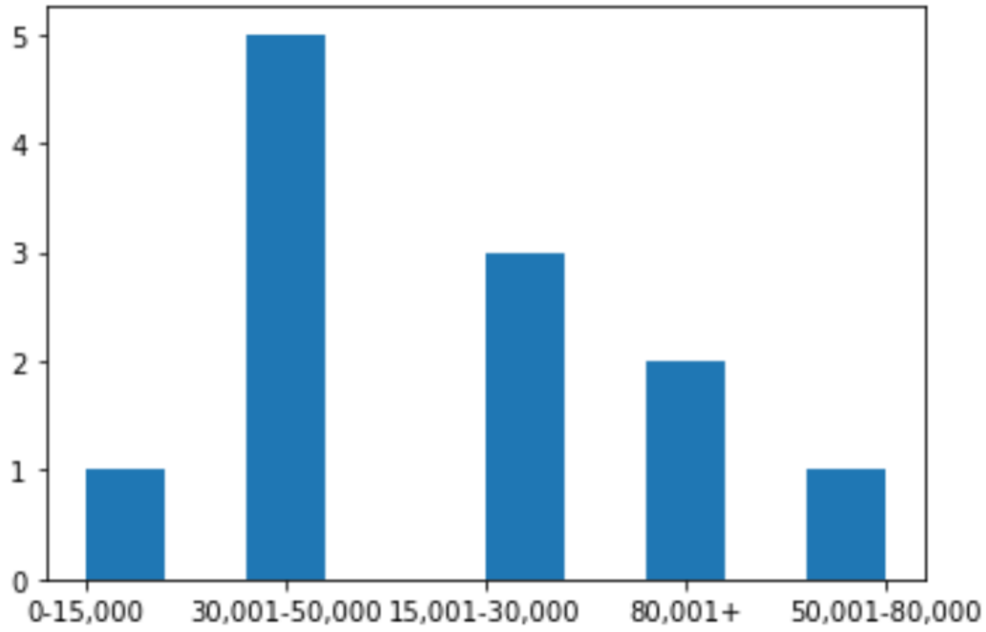
In [5]:
```
labels = ['0-15,000','15,001-30,000','30,001-50,000','50,001-80,000','80,001+']
```

In [6]:
```
df['bins'] = pd.cut(df['salary'], bins=bins, labels=labels, include_lowest=True)
df.sort_values(by='bins')
#df.bins.hist(grid=False)
```


[Jesús Rogel-Salazar, 2017]

binning

```
In [7]: df.bins.hist(grid=False)
Out[7]: <AxesSubplot:>
```



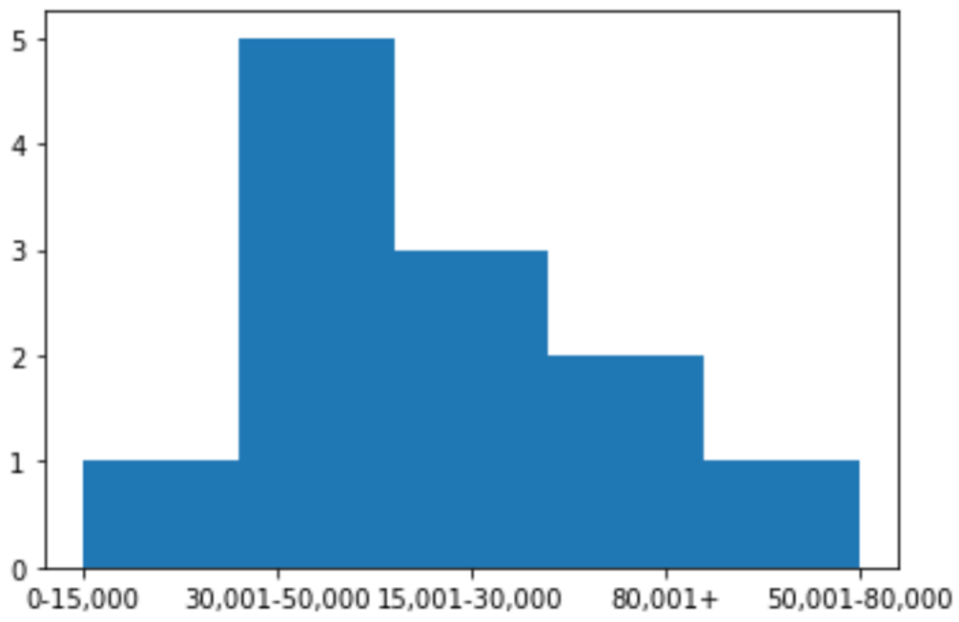
[Jesús Rogel-Salazar, 2017]

binning

```
In [8]: import matplotlib.pyplot as plt

        plt.hist(df['bins'], bins=5)

Out[8]: (array([1., 5., 3., 2., 1.]),
         array([0. , 0.8, 1.6, 2.4, 3.2, 4. ]),
         <BarContainer object of 5 artists>)
```



[Jesús Rogel-Salazar, 2017]

normalized data (z-score)

$$v' = \frac{v - \mu_A}{\sigma_A}$$

μ_A คือค่าเฉลี่ย

σ_Aคือค่าส่วนเบี่ยงเบนมาตราฐาน

Han, J. 2011

https://github.com/prasertcbs/tutorial/raw/master/mpg.csv

normalized data (z-score)

In [1]:
```
import pandas as pd
%matplotlib inline
print(f'pandas version = {pd.__version__}')
```

```
pandas version = 0.22.0
```

In [2]:
```
url='https://github.com/prasertcbs/tutorial/raw/master/mpg.csv'
df=pd.read_csv(url)
df[:10]
```

Out[2]:

	manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
0	audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	compact
1	audi	a4	1.8	1999	4	manual(m5)	f	21	29	p	compact
2	audi	a4	2.0	2008	4	manual(m6)	f	20	31	p	compact
3	audi	a4	2.0	2008	4	auto(av)	f	21	30	p	compact
4	audi	a4	2.8	1999	6	auto(l5)	f	16	26	p	compact
5	audi	a4	2.8	1999	6	manual(m5)	f	18	26	p	compact
6	audi	a4	3.1	2008	6	auto(av)	f	18	27	p	compact
7	audi	a4 quattro	1.8	1999	4	manual(m5)	4	18	26	p	compact
8	audi	a4 quattro	1.8	1999	4	auto(l5)	4	16	25	p	compact
9	audi	a4 quattro	2.0	2008	4	manual(m6)	4	20	28	p	compact

In [3]:
```
x_bar=df.cty.mean()
sd=df.cty.std()

print(f'mean={x_bar}, sd={sd}')
```

```
mean=16.858974358974358, sd=4.255945678889395
```


[Jesús Rogel-Salazar, 2017]

normalized data (z-score)

```
In [4]: (18-x_bar)/sd

Out[4]: 0.26810155183263457

In [5]: (df.cty-x_bar)/sd

Out[5]: 0     0.268102
        1     0.972998
        2     0.738032
        3     0.972998
        4    -0.201829
        5     0.268102
        6     0.268102
        7     0.268102
        8    -0.201829
```


[Jesús Rogel-Salazar, 2017]

normalized data (z-score)

In [6]:
```
df['z_cty']=(df.cty-x_bar)/sd
```

In [21]:
```
df.head()
```

Out[21]:

	manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class	z_cty	cty_z	hwy_z	displ_z
0	audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	compact	0.268102	0.268102	0.933696	-1.294000
1	audi	a4	1.8	1999	4	manual(m5)	f	21	29	p	compact	0.972998	0.972998	0.933696	-1.294000
2	audi	a4	2.0	2008	4	manual(m6)	f	20	31	p	compact	0.738032	0.738032	1.269569	-1.139196
3	audi	a4	2.0	2008	4	auto(av)	f	21	30	p	compact	0.972998	0.972998	1.101633	-1.139196
4	audi	a4	2.8	1999	6	auto(l5)	f	16	26	p	compact	-0.201829	-0.201829	0.429888	-0.519982

In [8]:
```
df.z_cty.mean()
```

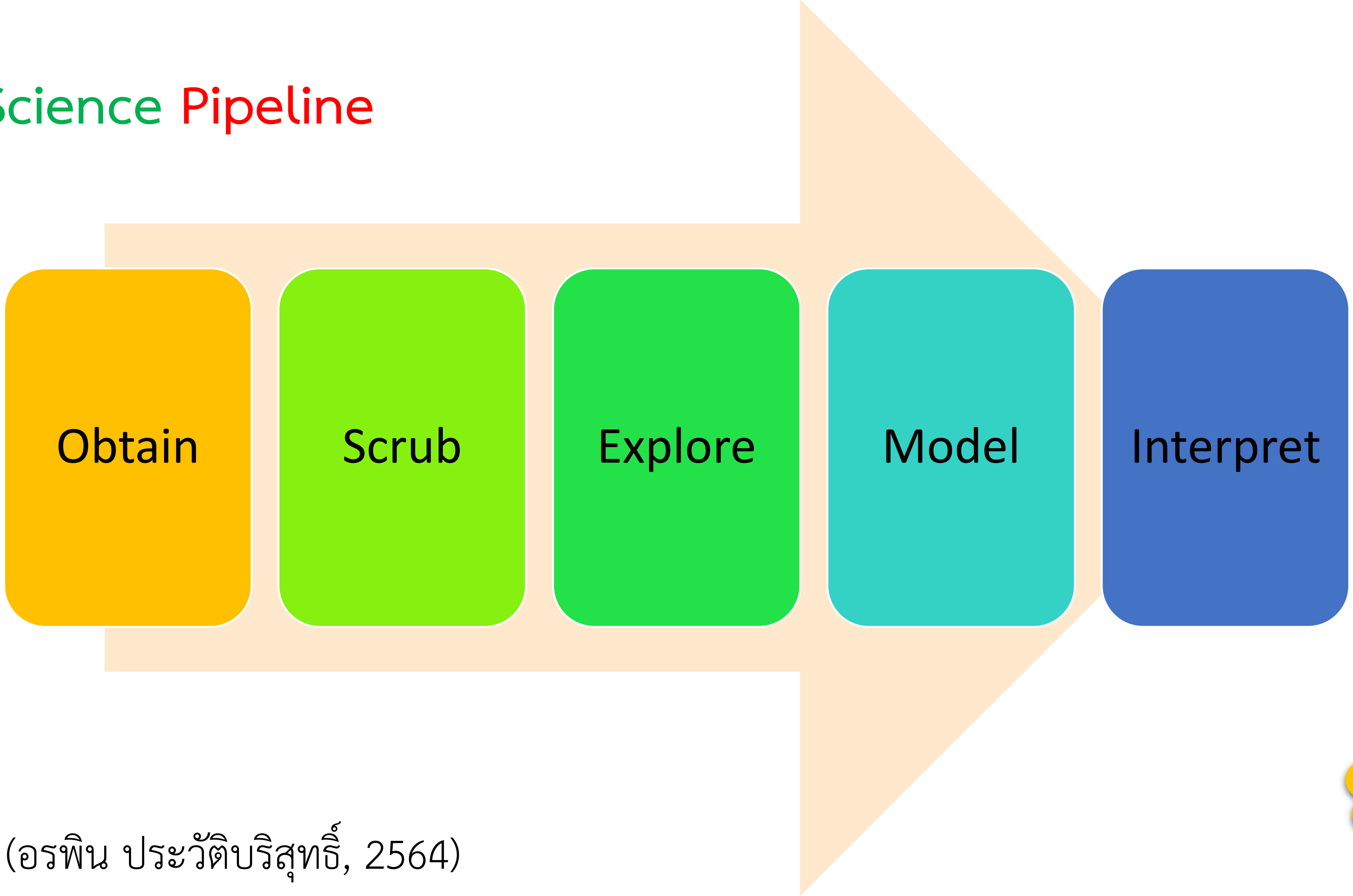
Out[8]: 3.036507417778206e-16

In [9]:
```
df.z_cty.std()
```

Out[9]: 1.0

ขั้นตอนการทำ Data Science

Data Science Pipeline



(อรพิน ประวัติบริสุทธิ์, 2564)

CS

Model

Model Evaluation

- training dataset and testing dataset

Classification method

- Simple linear regression
- Naïve bayes
- Decision tree
- Neural Network
- Support Vector machine

Clustering Method

Han, J. 2011

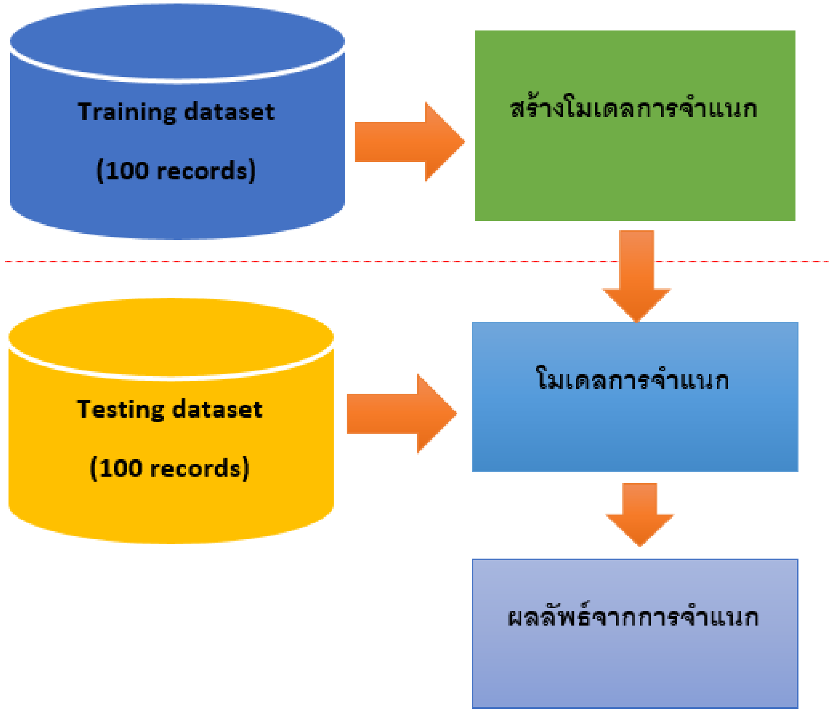
Model Evaluation

รูปแบบการแบ่งข้อมูลเพื่อทำการทดสอบ

- Self-Consistency Test
- Spilt Test
- Cross Validation Test

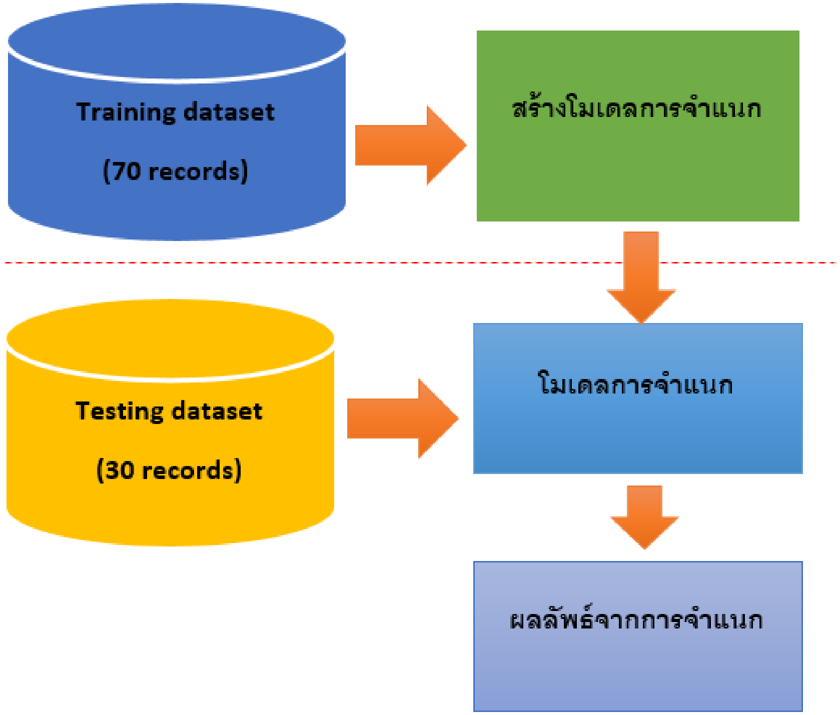
Han, J. 2011

Self-Consistency Test



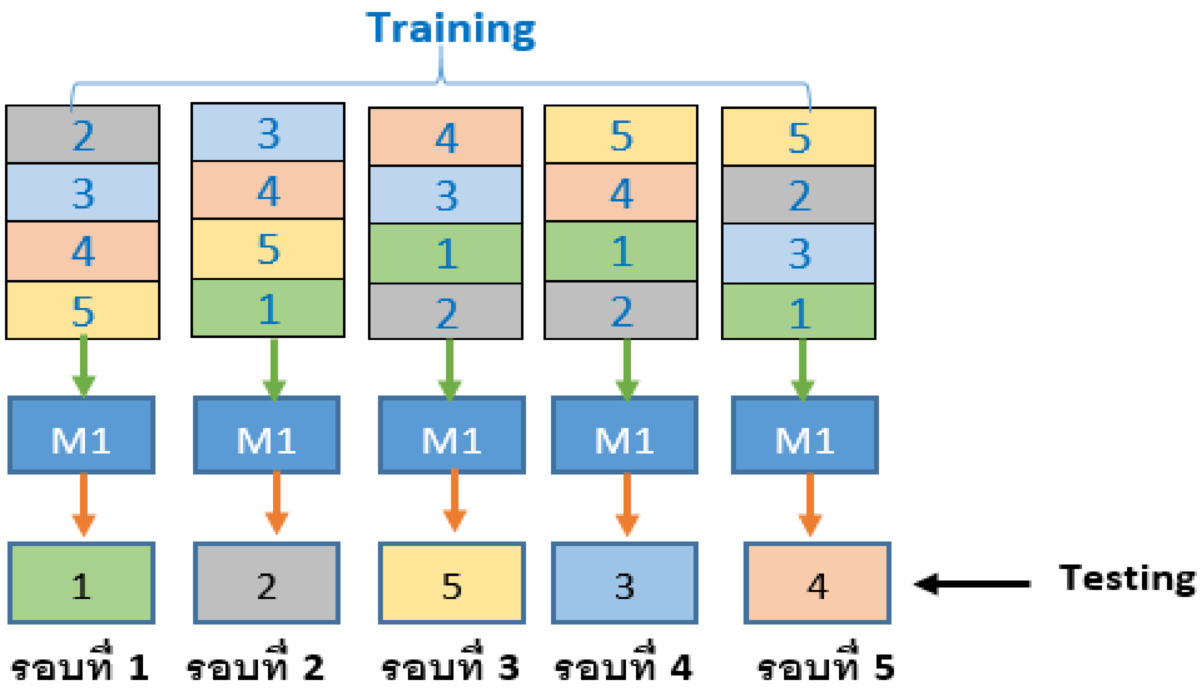
Han, J. 2011

Spilt Test



Han, J. 2011

Cross Validation Test



Han, J. 2011

การแบ่ง DataFrame ออกเป็น training และ test datasets

In [1]:
```python
import pandas as pd
import numpy as np
```

In [21]:
```python
url = 'https://github.com/prasertcbs/tutorial/raw/master/mtcars.csv'
# df = pd.read_csv(url, nrows=10)
df = pd.read_csv(url)
df
```

Out[21]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
2	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
8	Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
9	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
10	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4

https://github.com/prasertcbs/tutorial/raw/master/mtcars.csv

[Jesús Rogel-Salazar, 2017]

การแบ่ง DataFrame ออกเป็น training และ test datasets

```
In [18]: df.shape
Out[18]: (32, 12)
```


[Jesús Rogel-Salazar, 2017]

การแบ่ง DataFrame ออกเป็น training และ test datasets (70:30)

In [24]:
```
train_ratio = .7
df_train = df[:int(df.shape[0] * train_ratio)]
df_train
```

Out[24]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb	uid
20	Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1	-1913866085
28	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4	-1901534138
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4	-1738417093
26	Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2	-1735173775
12	Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3	-1714342737
21	Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	0	3	2	-1538944093
5	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1	-1330301436

In [28]:
```
df_train.shape
```


[Jesús Rogel-Salazar, 2017]

การแบ่ง DataFrame ออกเป็น training และ test datasets (70:30)

```
In [29]: df_test = df[int(df.shape[0] * train_ratio):]
         df_test
```

```
In [30]: df_test.shape
```


[Jesús Rogel-Salazar, 2017]

การแบ่ง DataFrame ออกเป็น training และ test datasets

Use index to randomly split dataset into training and test datasets

In [1]:
```
import pandas as pd
```

In [2]:
```
url = 'https://github.com/prasertcbs/tutorial/raw/master/msleep.csv'
df = pd.read_csv(url, nrows=10)
df
```

Out[2]:

	name	genus	vore	order	conservation	sleep_total	sleep_rem	sleep_cycle	awake	brainwt	bodywt
0	Cheetah	Acinonyx	carni	Carnivora	lc	12.1	NaN	NaN	11.9	NaN	50.000
1	Owl monkey	Aotus	omni	Primates	NaN	17.0	1.8	NaN	7.0	0.01550	0.480
2	Mountain beaver	Aplodontia	herbi	Rodentia	nt	14.4	2.4	NaN	9.6	NaN	1.350
3	Greater short-tailed shrew	Blarina	omni	Soricomorpha	lc	14.9	2.3	0.133333	9.1	0.00029	0.019
4	Cow	Bos	herbi	Artiodactyla	domesticated	4.0	0.7	0.666667	20.0	0.42300	600.000
5	Three-toed sloth	Bradypus	herbi	Pilosa	NaN	14.4	2.2	0.766667	9.6	NaN	3.850
6	Northern fur seal	Callorhinus	carni	Carnivora	vu	8.7	1.4	0.383333	15.3	NaN	20.490
7	Vesper mouse	Calomys	NaN	Rodentia	NaN	7.0	NaN	NaN	17.0	NaN	0.045
8	Dog	Canis	carni	Carnivora	domesticated	10.1	2.9	0.333333	13.9	0.07000	14.000
9	Roe deer	Capreolus	herbi	Artiodactyla	lc	3.0	NaN	NaN	21.0	0.09820	14.800

https://github.com/prasertcbs/tutorial/raw/master/msleep.csv

[Jesús Rogel-Salazar, 2017]

```
In [3]: train_ratio = .7

In [4]: df_train = df.sample(frac=train_ratio)
        df_train
```

Out[4]:

	name	genus	vore	order	conservation	sleep_total	sleep_rem	sleep_cycle	awake	brainwt	bodywt
7	Vesper mouse	Calomys	NaN	Rodentia	NaN	7.0	NaN	NaN	17.0	NaN	0.045
2	Mountain beaver	Aplodontia	herbi	Rodentia	nt	14.4	2.4	NaN	9.6	NaN	1.350
4	Cow	Bos	herbi	Artiodactyla	domesticated	4.0	0.7	0.666667	20.0	0.42300	600.000
1	Owl monkey	Aotus	omni	Primates	NaN	17.0	1.8	NaN	7.0	0.01550	0.480
3	Greater short-tailed shrew	Blarina	omni	Soricomorpha	lc	14.9	2.3	0.133333	9.1	0.00029	0.019
5	Three-toed sloth	Bradypus	herbi	Pilosa	NaN	14.4	2.2	0.766667	9.6	NaN	3.850
9	Roe deer	Capreolus	herbi	Artiodactyla	lc	3.0	NaN	NaN	21.0	0.09820	14.800

[Jesús Rogel-Salazar, 2017]

```
In [5]: df_train.index

Out[5]: Int64Index([7, 2, 4, 1, 3, 5, 9], dtype='int64')

In [6]: df.index.isin(df_train.index)

Out[6]: array([False,  True,  True,  True,  True,  True, False,  True, False,
                True])
```


[Jesús Rogel-Salazar, 2017]

```
In [7]: ~df.index.isin(df_train.index)

Out[7]: array([ True, False, False, False, False, False,  True, False,  True,
        False])

In [8]: df_test = df[~df.index.isin(df_train.index)]
        df_test

Out[8]:
```

	name	genus	vore	order	conservation	sleep_total	sleep_rem	sleep_cycle	awake	brainwt	bodywt
0	Cheetah	Acinonyx	carni	Carnivora	lc	12.1	NaN	NaN	11.9	NaN	50.00
6	Northern fur seal	Callorhinus	carni	Carnivora	vu	8.7	1.4	0.383333	15.3	NaN	20.49
8	Dog	Canis	carni	Carnivora	domesticated	10.1	2.9	0.333333	13.9	0.07	14.00

[Jesús Rogel-Salazar, 2017]

In [54]:
```
df_train.index
```
Out[54]:
```
Int64Index([12, 1, 6, 9, 13, 3, 5, 7, 16, 8, 11, 15], dtype='int64')
```
In [55]:
```
df.index.isin(df_train.index)
```
Out[55]:
```
array([False,  True, False,  True, False,  True,  True,  True,  True,
        True, False,  True,  True,  True, False,  True,  True, False,
       False, False])
```
In [56]:
```
~df.index.isin(df_train.index)
```
Out[56]:
```
array([ True, False,  True, False,  True, False, False, False, False,
       False,  True, False, False, False,  True, False, False,  True,
        True,  True])
```
In [57]:
```
df_test = df[~df.index.isin(df_train.index)]
```
In [58]:
```
df_test
```

อ้างอิง

- Jesús Rogel-Salazar.DATA SCIENCE AND ANALYTICS WITH PYTHON CRC Press Taylor & Francis Group, 2017.
- Jake VanderPlas. Python Data Science Handbook . O'REILLY, 2017.